

8.2 Scaling



Core skills you'll master in this section:

- ▶ The determinant of the transformation matrix: the area/volume scaling ratio.
- ▶ Squeeze transformation: understand scaling under the condition that area remains unchanged.
- ▶ Scaling satisfies the commutative law of matrix multiplication.
- ▶ Negative scaling factors: scaling that includes a reflection.
- ▶ A scaling factor of 0: the geometric transformation includes a dimension reduction, is not invertible, and loses information.

Among geometric transformations, scaling is one of the most fundamental operations. It can be efficiently implemented through matrix multiplication, and can handle various cases such as uniform scaling, non-uniform scaling, successive scaling, and negative-ratio scaling. This section covers scaling systematically – we'll start with the 2D plane and then extend to 3D space.

Uniform Scaling in the Plane

Uniform scaling means enlarging or shrinking a shape by the same ratio in all directions. The linear transformation matrix for uniform scaling in the plane has the following form

$$S = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = sI$$

If $s > 1$, this means enlargement (as in Figure 1 (a)); if $0 < s < 1$, this means shrinking (as in Figure 1 (b)). This book introduced this kind of diagonal matrix (with identical main-diagonal entries) earlier – it acts like scalar multiplication of a matrix.

The square matrix's determinant is s^2 ; this means that after the transformation, the geometric shape's area changes by a factor of s^2 .

the square matrix S 's column vectors form an orthogonal basis.

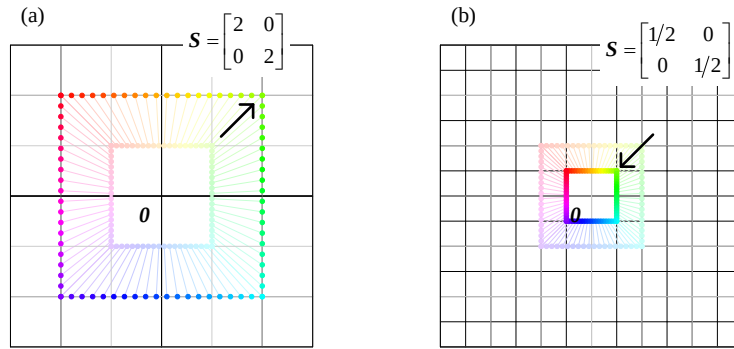


Figure 1. Uniform Scaling, 2D Plane

Applying uniform scaling to the vector $[x_1, x_2]^T$ gives the new vector $[y_1, y_2]^T$:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_1 \\ sx_2 \end{bmatrix}$$

After scaling, the horizontal and vertical axes are scaled by the same ratio, so the shape is simply enlarged or shrunk as a whole.

⚠ Note that the origin's position doesn't change before or after scaling.

It's easy to see that Figure 1 (a) and (b) are inverse operations of each other, that is,

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

As mentioned earlier in the book, if a diagonal square matrix has no zero entries on its main diagonal, its inverse is also a diagonal matrix, whose main-diagonal entries are the reciprocals of the corresponding entries in the original matrix. If the diagonal square matrix has one or more zero entries on its main diagonal, the matrix is not invertible.

Non-Uniform Scaling

Non-uniform scaling (in the plane) means the horizontal and vertical directions are scaled by different ratios; the transformation matrix is:

$$S = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$$

Here, s_1 and s_2 are both greater than 0, and $s_1 \neq s_2$. If either scaling coefficient equals 1, that means there's no transformation in the corresponding direction.

Please analyze on your own the two panels in Figure 2.

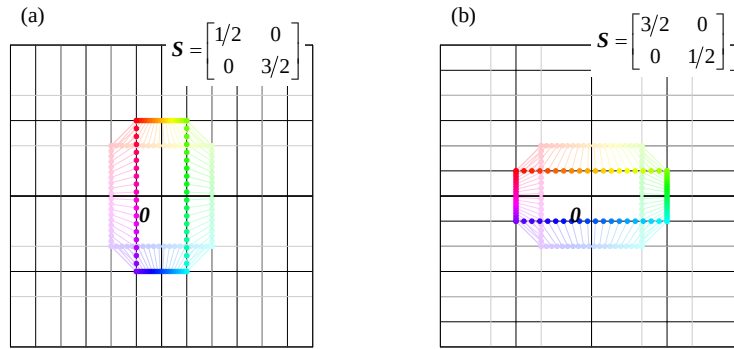


Figure 2. Non-Uniform Scaling, 2D Plane

The book introduced the scaling effect of diagonal matrices earlier – let's briefly review it below.

For example, given the 2×2 matrix $\mathbf{A}$, the matrix multiplication $\mathbf{A} @ \mathbf{S}$ is equivalent to scaling $\mathbf{A}$'s column vectors in turn by the diagonal matrix's main-diagonal entries, that is

$$\mathbf{AS} = [\mathbf{a}_1 \quad \mathbf{a}_2] \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix} = [s_1 \mathbf{a}_1 \quad s_2 \mathbf{a}_2]$$

Whereas the matrix multiplication $\mathbf{S} @ \mathbf{A}$ is equivalent to scaling $\mathbf{A}$'s row vectors in turn by the diagonal matrix's main-diagonal entries, that is

$$\mathbf{SA} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix} \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \end{bmatrix} = \begin{bmatrix} s_1 \mathbf{a}^{(1)} & s_2 \mathbf{a}^{(2)} \end{bmatrix}$$

As shown in Figure 3, $\det(\mathbf{S}) \neq 0$, so the scaling matrix $\mathbf{S}$ has an inverse, that is

$$\mathbf{S}^{-1} = \begin{bmatrix} 1/s_1 & 0 \\ 0 & 1/s_2 \end{bmatrix}$$

If s_1 or s_2 equals 0, the scaling matrix $\mathbf{S}$ is not invertible, and the transformation cannot be undone.

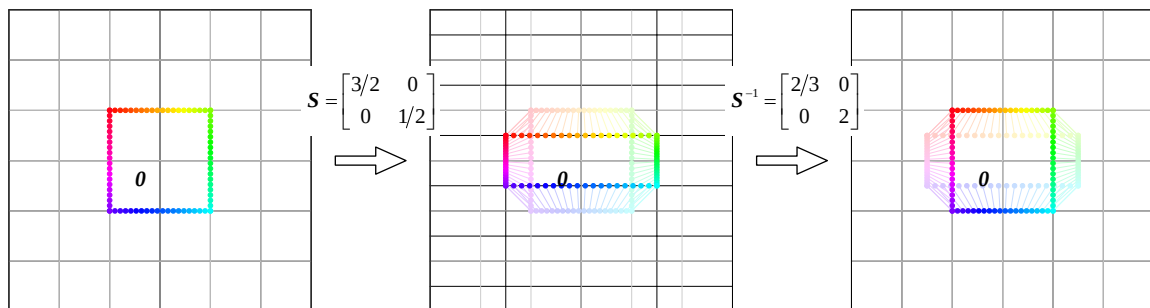


Figure 3. The Inverse of Scaling

Since the scaling matrix $\mathbf{S} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$ is a diagonal matrix, its determinant is easy to compute

$$\det(\mathbf{S}) = s_1 s_2$$

If $|\det(\mathbf{S})| > 1$, that means the planar shape's area is enlarged; $0 < |\det(\mathbf{S})| < 1$ means the area is shrunk. $\det(\mathbf{S}) = 0$ means a dimension reduction occurs, information is lost, and the matrix is not invertible.

Horizontal and Vertical Scaling

When scaling occurs only in the horizontal direction, the corresponding transformation matrix is

$$\mathbf{S} = \begin{bmatrix} s & 0 \\ 0 & 1 \end{bmatrix}$$

This diagonal matrix's determinant is s , meaning the area is scaled by a factor of s after the geometric transformation. Figure 4 (a) and (b) give two examples of horizontal scaling.

When scaling occurs only in the vertical direction, the corresponding transformation matrix is

$$\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & s \end{bmatrix}$$

Figure 4 (c) and (d) give two examples of vertical scaling.

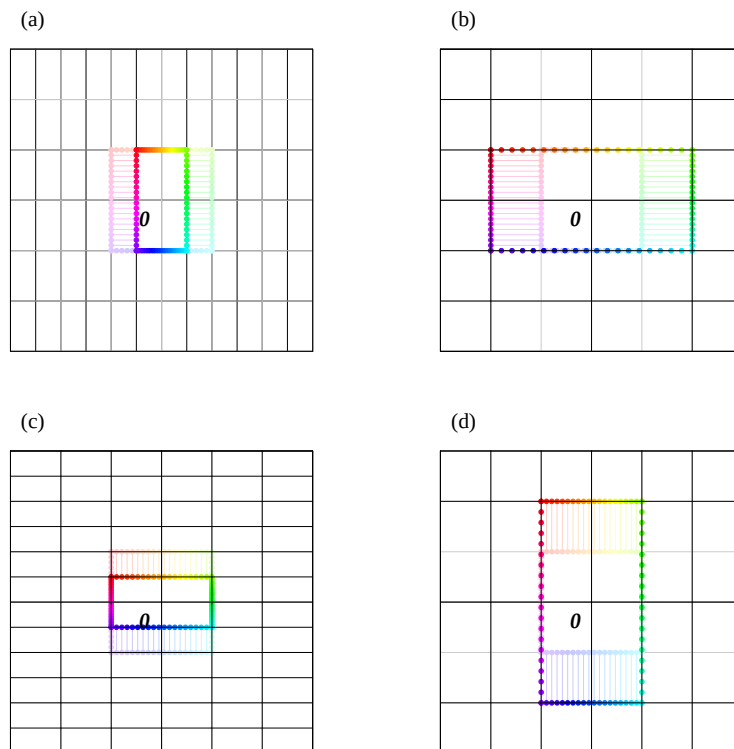


Figure 4. Horizontal and Vertical Scaling, 2D Plane

Squeeze

Squeeze is a special kind of non-uniform scaling. In the plane, a squeeze means the scaling coefficient in one direction is less than 1 while the scaling coefficient in the other direction is greater than 1, with the two multiplying to 1, satisfying $s_1 s_2 = 1$ — that is, the determinant is 1.

For example, if $0 < s_1 < 1$ and $s_2 > 1$, then the shape is compressed in the x_1 direction and stretched in the x_2 direction, but the total area stays unchanged.

Figure 5 (a), $s_1 = 0.5$ and $s_2 = 2$, meaning the horizontal direction is scaled down by a factor of 0.5 while the vertical direction is scaled up by a factor of 2, and the total area stays unchanged.

Please analyze on your own Figure 5 (b).

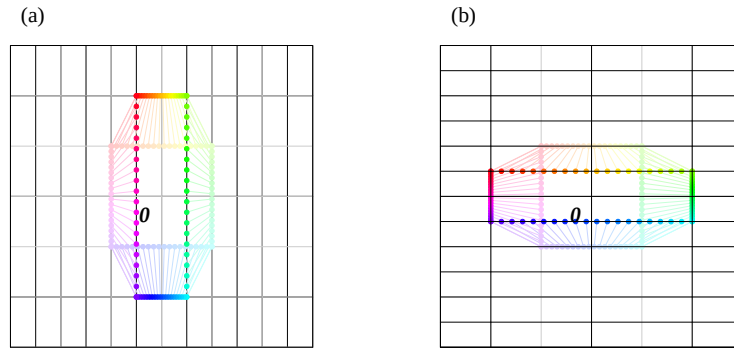


Figure 5. Squeeze

Successive Scaling

The book introduced earlier that, in general, matrix multiplication is not commutative; that is, for two matrices A and B , even if $A @ B$ and $B @ A$ both satisfy the rules of multiplication, generally $A @ B \neq B @ A$.

However, in a scaling transformation, the scaling matrix S is a diagonal matrix, and we can write the scaling matrix S as the product of two different matrices

$$S = \begin{bmatrix} s_1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 0 & s_2 \end{bmatrix} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 0 \\ 0 & s_2 \end{bmatrix} @ \begin{bmatrix} s_1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$$

tells us that, in special cases, matrix multiplication can satisfy the commutative law.

's first equation represents scaling in the x_2 direction first, then completing the scaling in the x_1 direction, that is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} s_1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 0 & s_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} s_1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ s_2 x_2 \end{bmatrix} = \begin{bmatrix} s_1 x_1 \\ s_2 x_2 \end{bmatrix}$$

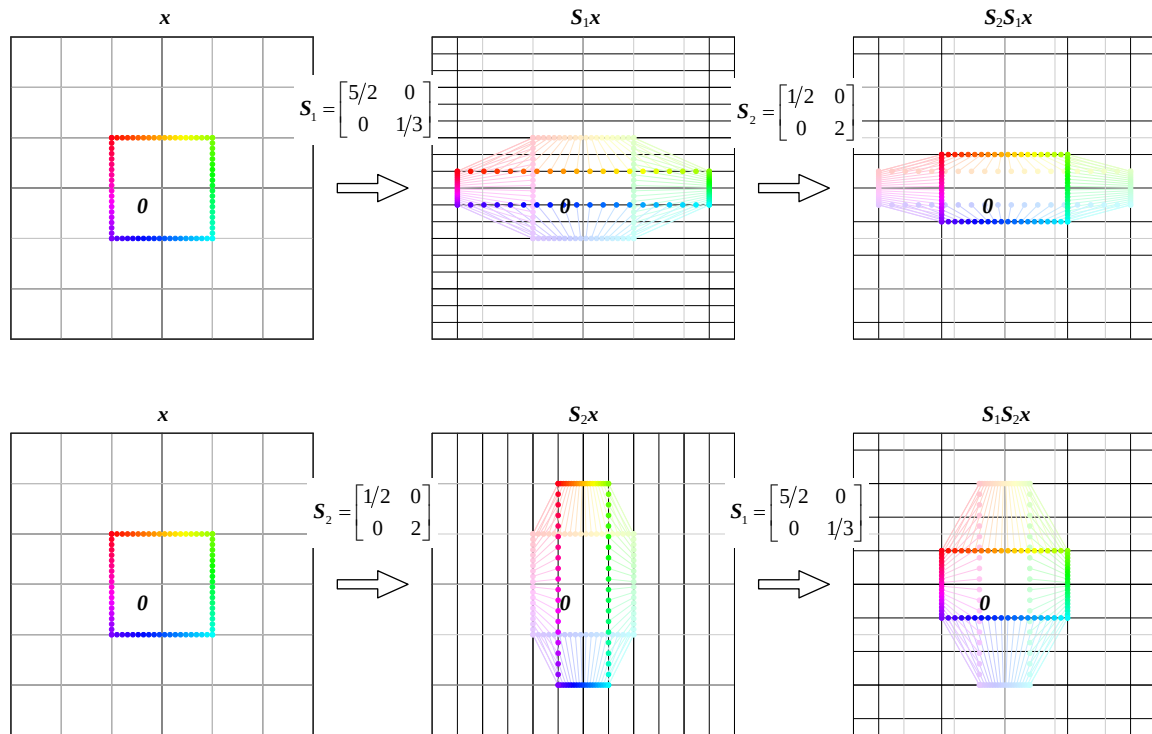
's second equation is the reverse of this — please work out the matrix multiplication yourself.

Geometrically, this shows that the order of scaling along the horizontal and vertical axes can be swapped.

Also, given any two scaling matrices S_1 and S_2 , we have

$$S_1 @ S_2 = S_2 @ S_1$$

This means the order in which multiple planar scaling operations are performed doesn't affect the final result — please take a look at Figure 6.

Figure 6. The Product of Two 2×2 Scaling Matrices Satisfies the Commutative Law

Negative Scaling Coefficients

If a scaling coefficient is negative, that means a flip occurs in the corresponding direction. In other words, negative-ratio scaling is equivalent to a composite transformation of “scaling + reflection.”

Figure 7 (a) corresponds to the following matrix multiplication

$$\begin{bmatrix} 2 & 0 \\ 0 & -1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The expression above also tells us that the order of these two geometric operations doesn't affect the result.

Please analyze on your own Figure 7 (b).

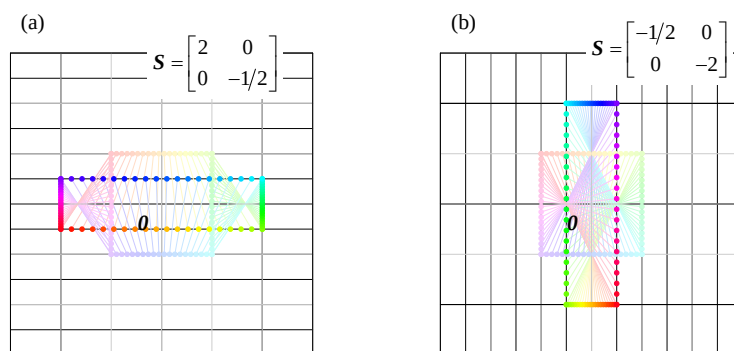


Figure 7. Scaling Factors Include Negative Values

Zero Scaling Coefficients

If a scaling coefficient is 0, that means a projection occurs in the corresponding direction. In this case, the geometric transformation is equivalent to a composite transformation of “scaling + projection.”

Figure 8 corresponds to the following operation

$$\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

All points are projected onto the vertical axis, so information is lost; the determinant is 0, which also means the matrix is not invertible. As mentioned earlier in this section, if a diagonal square matrix has one or more zero entries on its main diagonal, the matrix is not invertible.

In this case, the transformation matrix S 's column vectors cannot form a basis, because the column vectors are linearly dependent.

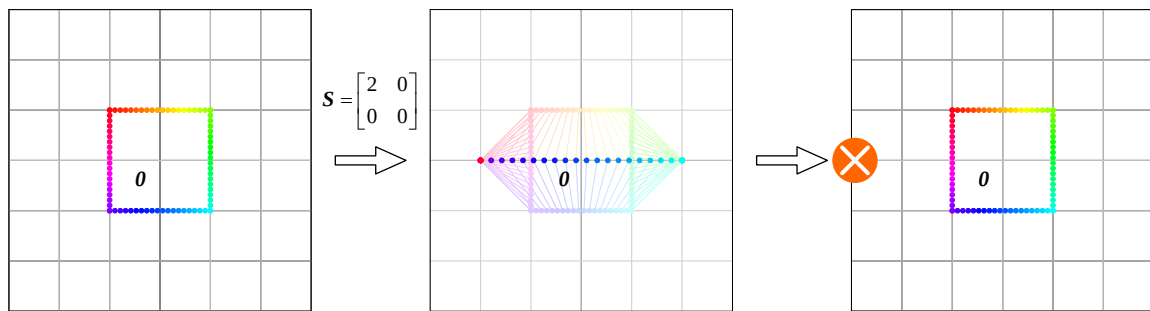


Figure 8. Irreversibly “Flattened”

Visualizing Planar Scaling

Code 1 is used to visualize planar scaling; the result is shown in Figure 9.

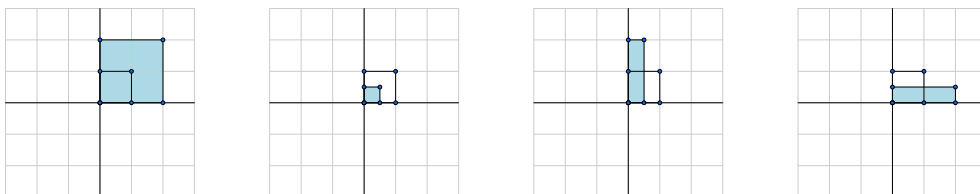


Figure 9. Visualizing Planar Scaling

Now let's go over Code 1's key statements.

a Uses `numpy.array()` to create the linear transformation matrix; try other 2×2 matrices on your own.

b Defines a list of vertices for the unit square, starting from the bottom-left corner and looping back to the starting point to close the shape. Each point is a 2D coordinate, forming an array of shape 5×2 . Note that each row vector represents a point in the plane.

c Defines unit vectors pointing in the positive horizontal-axis and positive vertical-axis directions, respectively.

d Uses the matrix A to perform the matrix multiplication, i.e., the linear transformation.

e is the visualization code. For example, the line `ax.plot(square_data[:,0], square_data[:,1], c = 'k', lw = '0.5', marker = 'o', markerfacecolor = '#0066FF', ls = '--')` draws the boundary of the original square. `square_data[:,0]` takes the horizontal coordinates of all the square's points, and `square_data[:,1]` takes their vertical coordinates.

`c='k'` sets the line color to black, and `lw='0.5'` sets the line width to 0.5.

`marker='o'` draws a small circle at each point, and `markerfacecolor='#0066FF'` sets the circles to blue.

`ls='--'` sets the line style to dashed.

Code 1. Visualizing Planar Scaling |  LA_Ch08_02_01.ipynb

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 线性变换矩阵
a A = np.array([[2, 0],
                [0, 2]])

## 单位正方形顶点数据
b square_data = np.array([[0, 0],
                          [1, 0],
                          [1, 1],
                          [0, 1],
                          [0, 0]])

## 定义单位向量 e1 和 e2
c e1 = np.array([[1], [0]])
  e2 = np.array([[0], [1]])

## 线性变换
d transformed_square = square_data @ A.T
  transformed_e1 = A @ e1
  transformed_e2 = A @ e2

## 可视化
fig, ax = plt.subplots(figsize=(6, 6))

e ax.plot(square_data[:,0], square_data[:,1],
          c = 'k', lw = '0.5',
            marker = 'o', markerfacecolor = '#0066FF', ls = '--')
  ax.plot(transformed_square[:,0], transformed_square[:,1],
          c = 'k', lw = '0.5',
            marker = 'o', markerfacecolor = '#0066FF', ls = '--')
  ax.fill(transformed_square[:,0], transformed_square[:,1],
          color='lightblue')

ax.set_aspect('equal'); ax.grid(True, c = '0.8')
ax.axhline(y = 0, c = 'k'); ax.axvline(x = 0, c = 'k')
ax.set_xlim(-3, 3); ax.set_ylim(-3, 3)
```

Uniform Scaling in 3D

In 3D space, the uniform scaling matrix is

$$\mathbf{S} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix}$$

Here, s is the scaling factor, and $s \neq 0$.

The corresponding coordinate transformation is

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} sx_1 \\ sx_2 \\ sx_3 \end{bmatrix} = s \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

In 3D space, uniform scaling applies the same scaling operation to all three dimensions through a single scaling factor s .

It's easy to compute that the determinant of the transformation matrix is s^3 , which means the volume changes by a factor of s^3 after the geometric transformation.

As shown in Figure 10, when $s > 1$, the unit cube is uniformly enlarged; as shown in Figure 11, when $0 < s < 1$, the unit cube is uniformly shrunk.

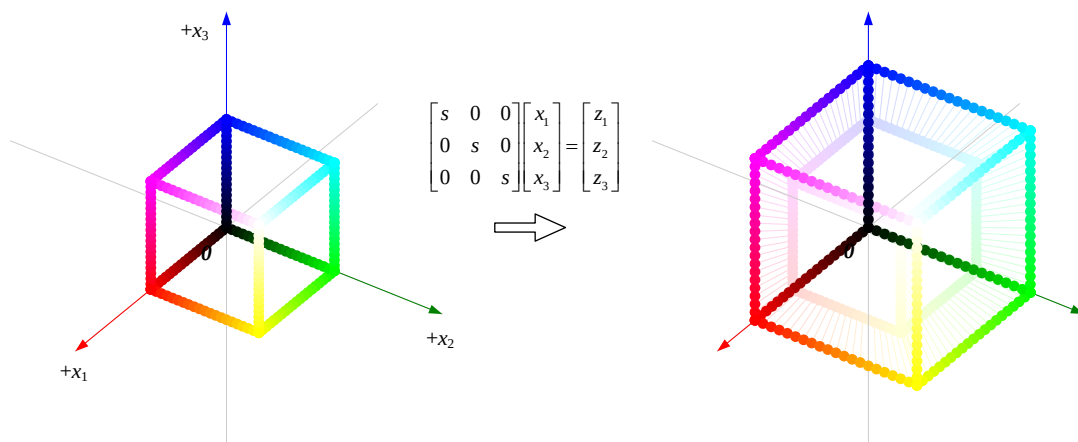
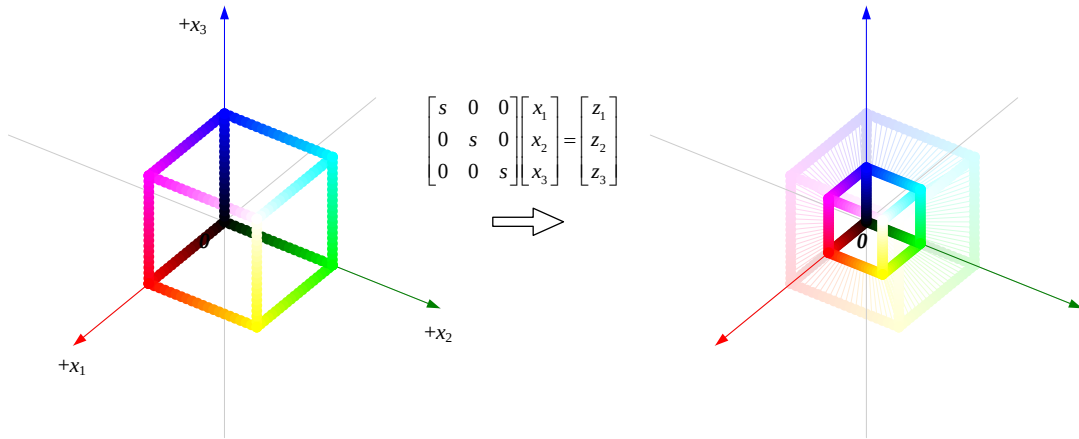


Figure 10. Uniform Enlargement ($s > 1$), 3D Space

Figure 11. Uniform Shrinking ($0 < s < 1$), 3D Space

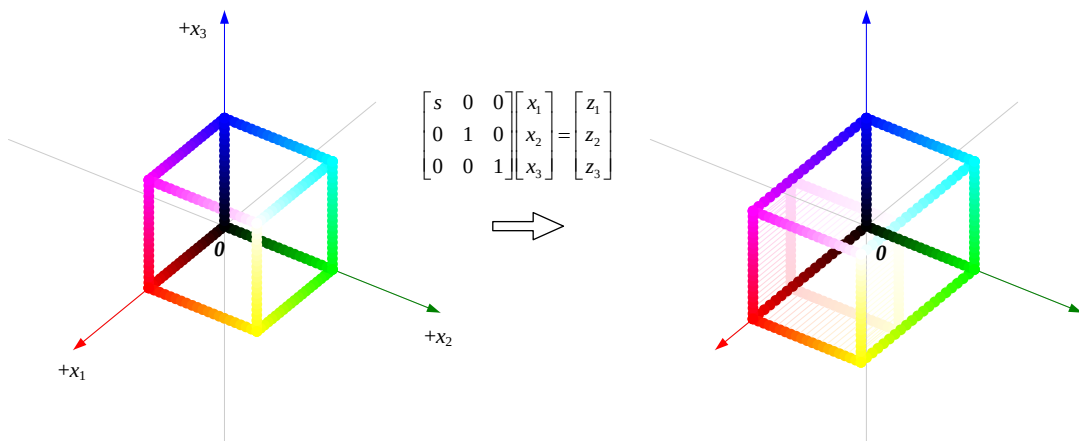
We can also decompose into scaling along each axis separately

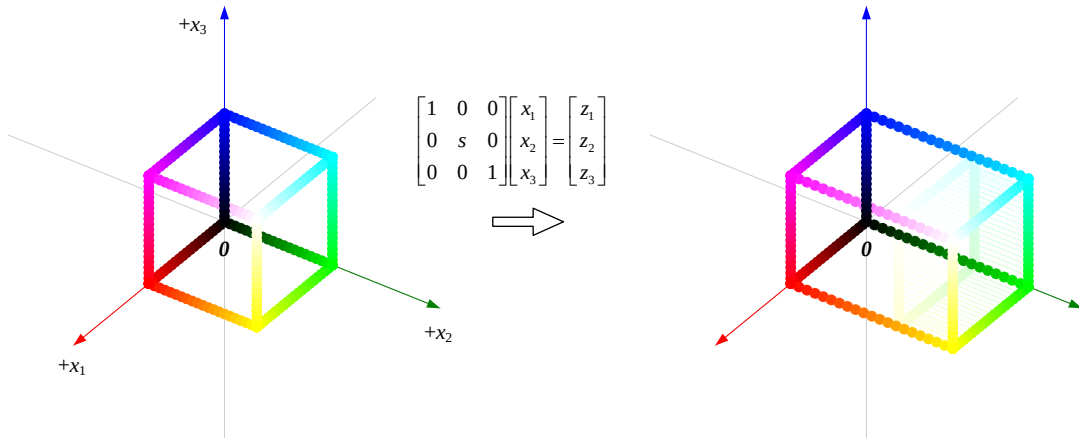
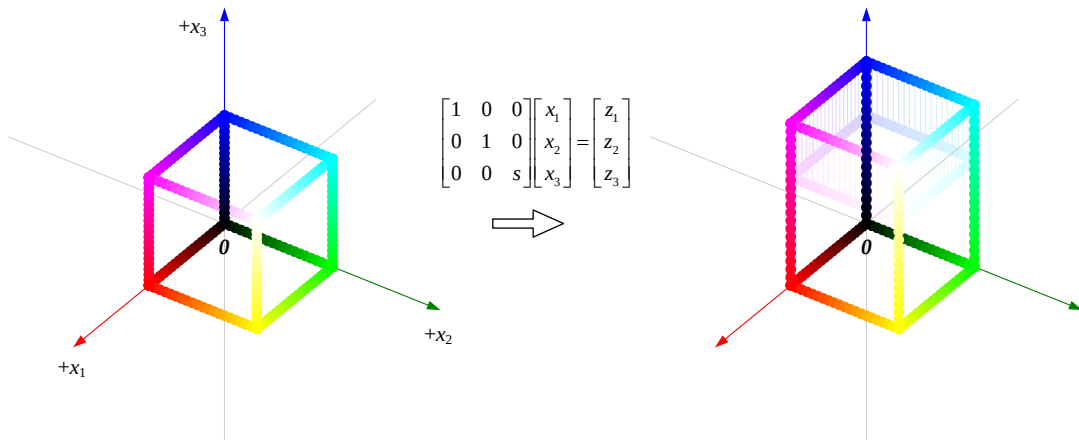
$$S_1 = \begin{bmatrix} s & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad S_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad S_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & s \end{bmatrix}$$

The three matrices above correspond to Figure 12, Figure 13, and Figure 14, respectively. The determinant of each of these three matrices is s , meaning the volume-scaling ratio caused by the geometric transformation is s in each case.

No matter what order these three diagonal matrices are multiplied in, the result is always , that is

$$S = S_1 S_2 S_3 = S_1 S_3 S_2 = S_2 S_1 S_3 = S_2 S_3 S_1 = S_3 S_1 S_2 = S_3 S_2 S_1$$

Figure 12. Scaling Along the x_1 Axis, 3D Space

Figure 13. Scaling Along the x_2 Axis, 3D SpaceFigure 14. Scaling Along the x_3 Axis, 3D Space

Non-Uniform Scaling in 3D

In 3D space, non-uniform scaling means applying different scaling factors to the x_1 , x_2 , and x_3 axes, respectively.

The scaling matrix extends to:

$$S = \begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & s_3 \end{bmatrix}$$

Here, s_1 controls the scaling in the x_1 direction, s_2 controls the scaling in the x_2 direction, and s_3 controls the scaling in the x_3 direction.

The corresponding coordinate transformation is

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & s_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} s_1 x_1 \\ s_2 x_2 \\ s_3 x_3 \end{bmatrix}$$

As shown in Figure 15, if the absolute value of a scaling factor is greater than 1, the shape is enlarged along the corresponding direction; if it's less than 1, the shape is shrunk along that direction.

Also, if a scaling factor is less than 0, that means a reflection is layered on top of the scaling; if a scaling factor is 0, that means a “dimension reduction” occurs and the shape is flattened, as shown in Figure 16.

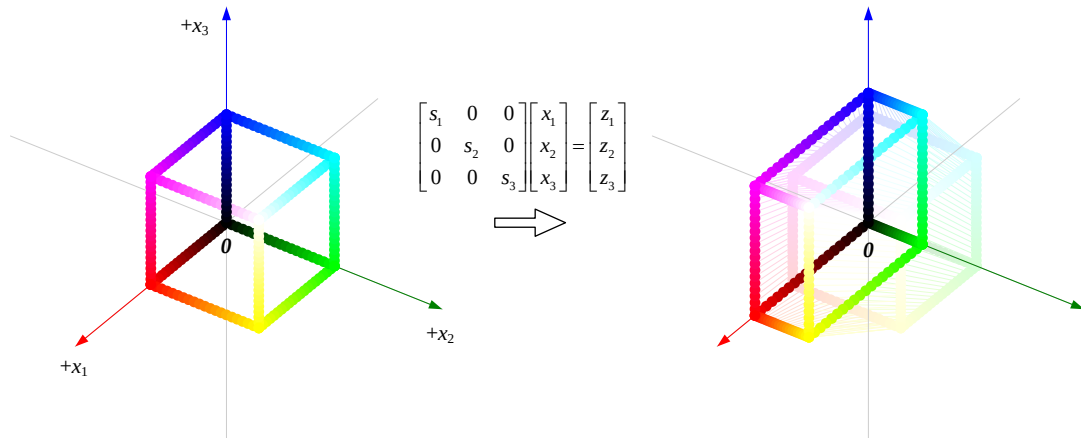


Figure 15. Non-Uniform Scaling, 3D Space

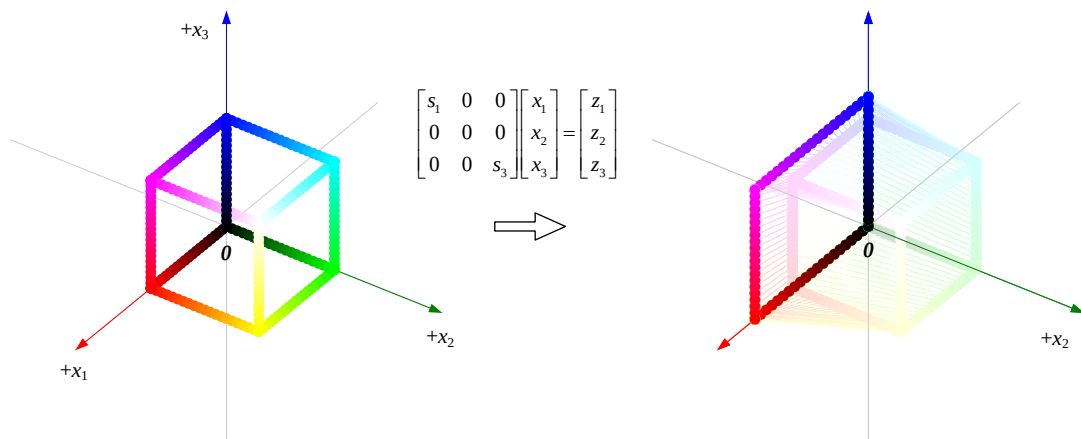


Figure 16. Scaling + Flattening, 3D Space



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. For a scaling transformation in the plane, if the horizontal axis is enlarged $2\times$ and the vertical axis is shrunk to $1/2$, what is the corresponding scaling matrix?

Q2. For a scaling transformation in 3D space, if the x_1 axis is enlarged $3\times$, the x_2 axis is shrunk to $1/3$, and the x_3 axis is unchanged, what is the corresponding scaling matrix?

$$\mathbf{S} = \begin{bmatrix} -2 & \\ & 2 \end{bmatrix}$$

Q3. Given the scaling matrix $\mathbf{S} = \begin{bmatrix} -2 & \\ & 2 \end{bmatrix}$, what do the coordinates of the point (3, 4) become after the scaling transformation?

$$\mathbf{S}_1 = \begin{bmatrix} 1 & \\ & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{S}_2 = \begin{bmatrix} 3 & \\ & 1 \end{bmatrix}$$

Q4. Given two scaling matrices $\mathbf{S}_1 = \begin{bmatrix} 1 & \\ & 2 \end{bmatrix}$ and $\mathbf{S}_2 = \begin{bmatrix} 3 & \\ & 1 \end{bmatrix}$, scale the point (1, 2) first by $\mathbf{S}_1$ and then by $\mathbf{S}_2$, and find the transformed coordinates. If the order of scaling is swapped, i.e., first $\mathbf{S}_2$ and then $\mathbf{S}_1$, what is the result?

Q5. If the scaling matrix's determinant $\det(\mathbf{S}) = 4$, what effect does this scaling transformation have on a geometric shape? And what does $\det(\mathbf{S}) = -4$ tell us?

Q6. Write a Python function `scale_2D(x1, x2, s1, s2)` that takes a point's coordinates (x1, x2) and scaling coefficients s1, s2 as input, and returns the transformed coordinates.

Q7. Given the 5 points (1,0), (0,1), (-1,0), (0,-1), (0,0), write Python code to scale these points by the ratios $s_1 = 2$ and $s_2 = 0.5$, and output the transformed set of points.

Q8. Use Matplotlib to draw a unit square whose four vertices are at (1,0), (0,1), (1,1), (0,0); scale the unit square by the ratios $s_1 = 2$ and $s_2 = 0.5$, and display both the original shape and the transformed shape.

Q9. Use next section's LA_08_03_01.ipynb to visualize 3D scaling.